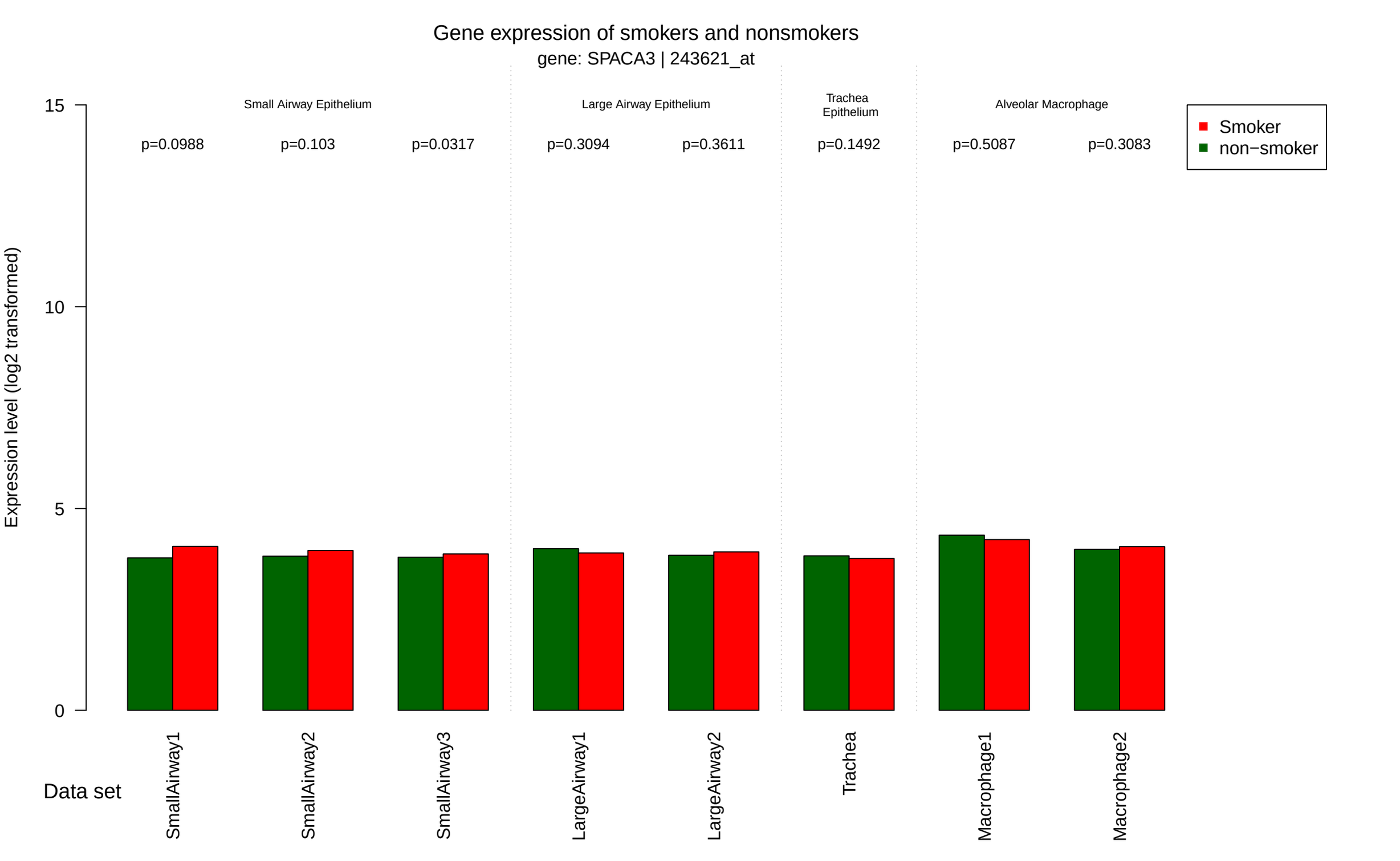
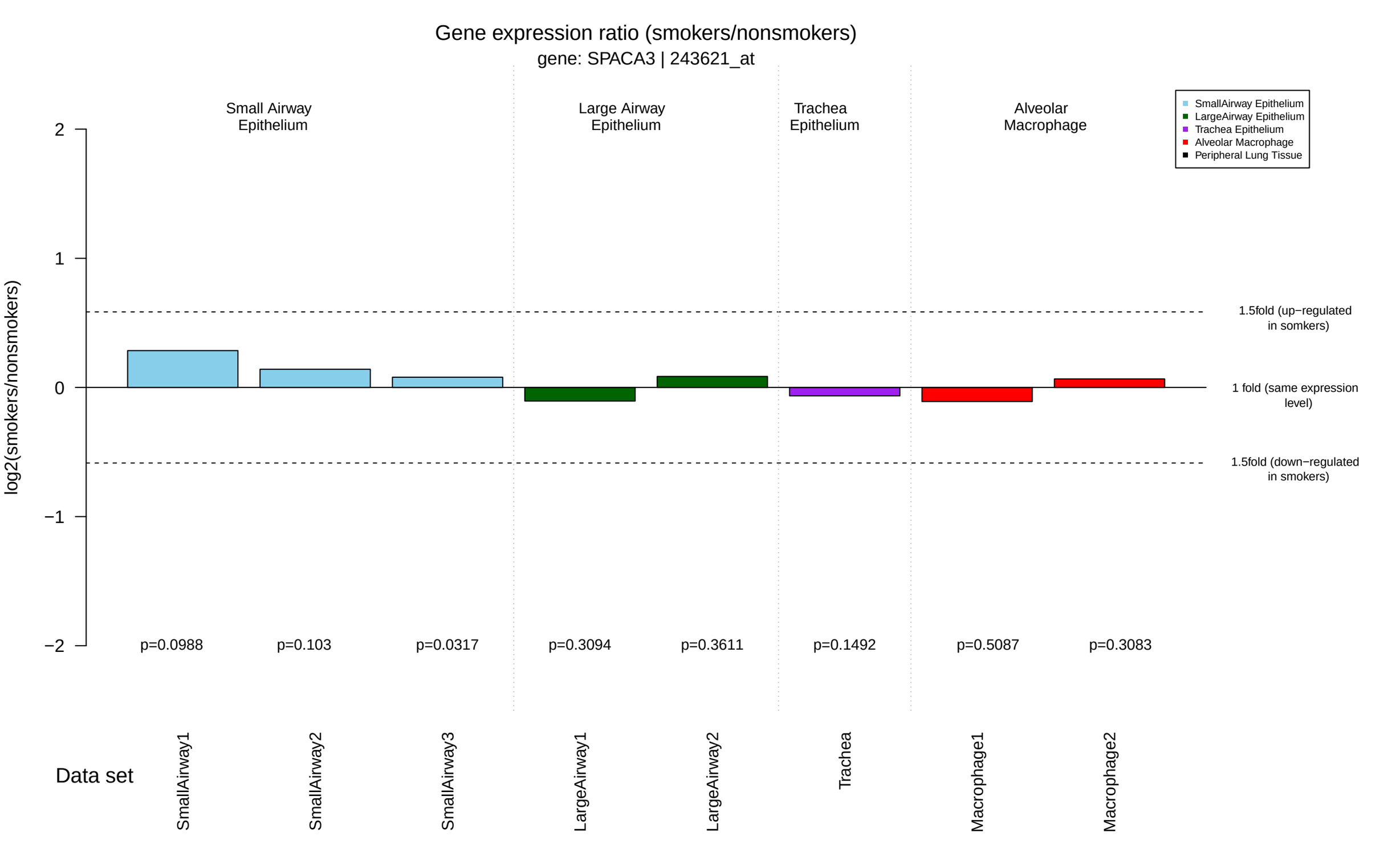
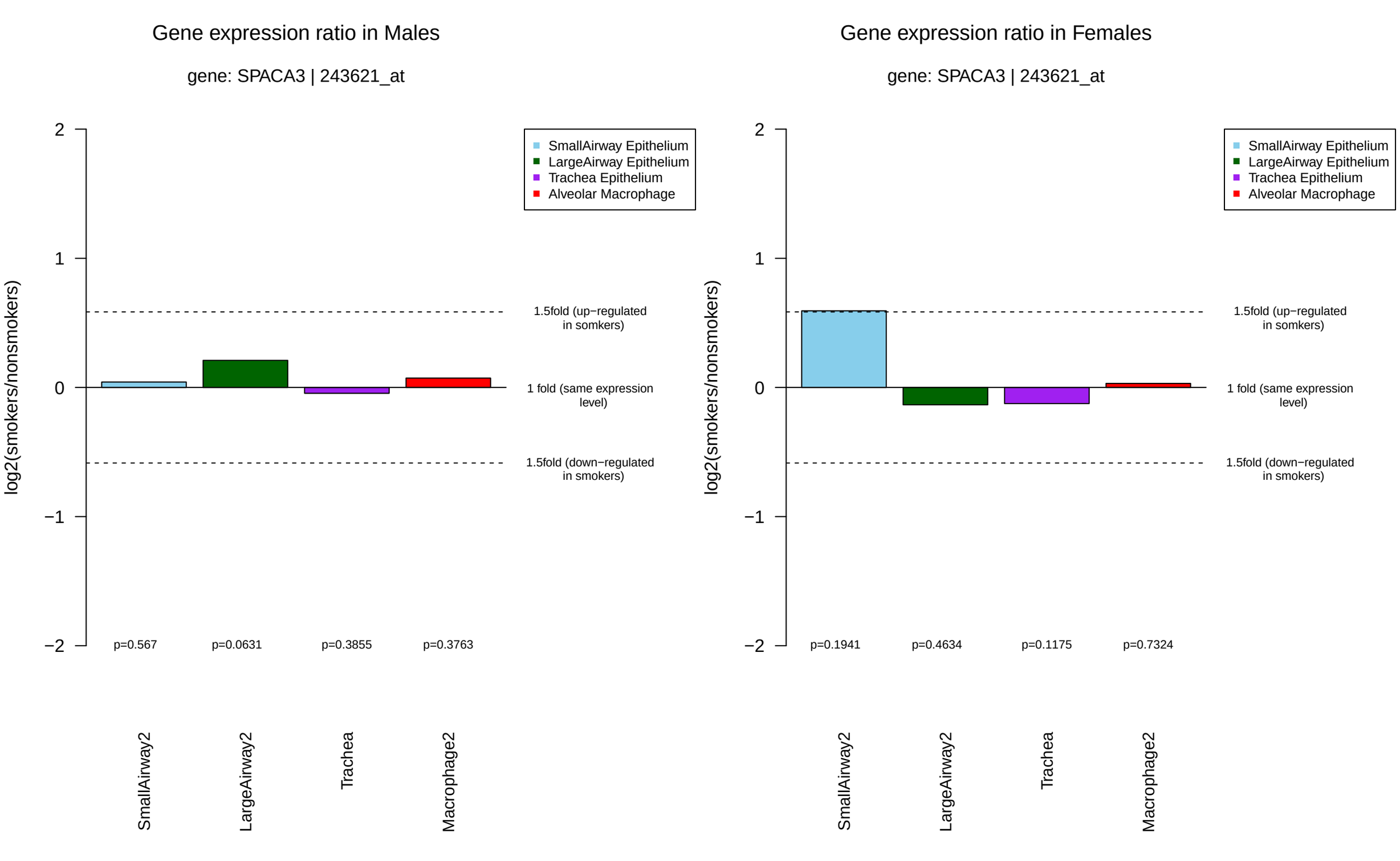
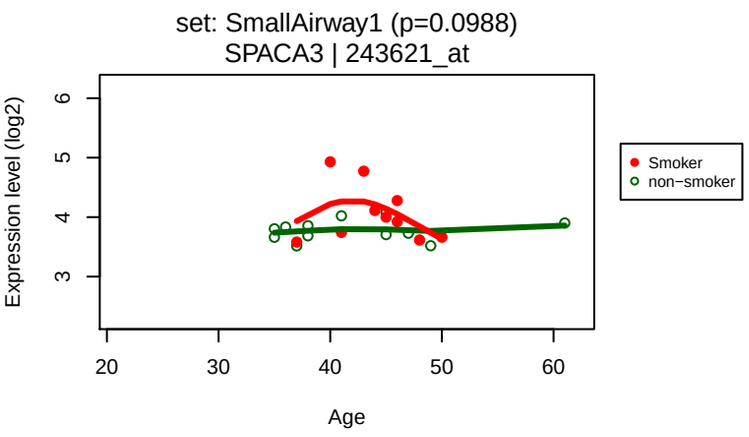
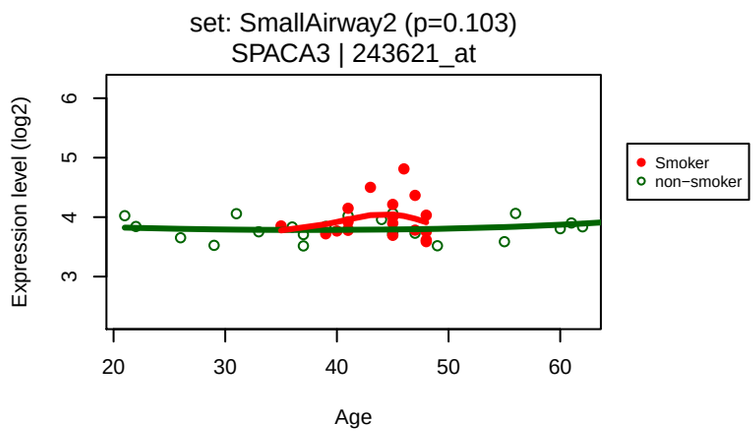
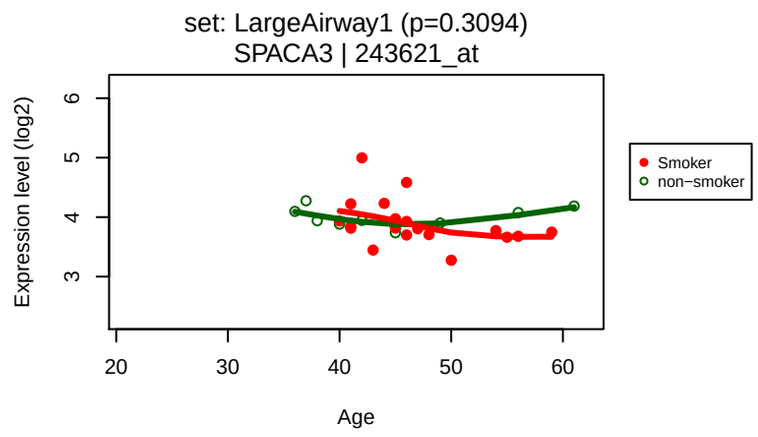

Gene expression of smokers and nonsmokers
gene: SPACA3 | 243621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0988
p=0.103
p=0.0317
p=0.3094
p=0.3611
p=0.1492
p=0.5087
p=0.3083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

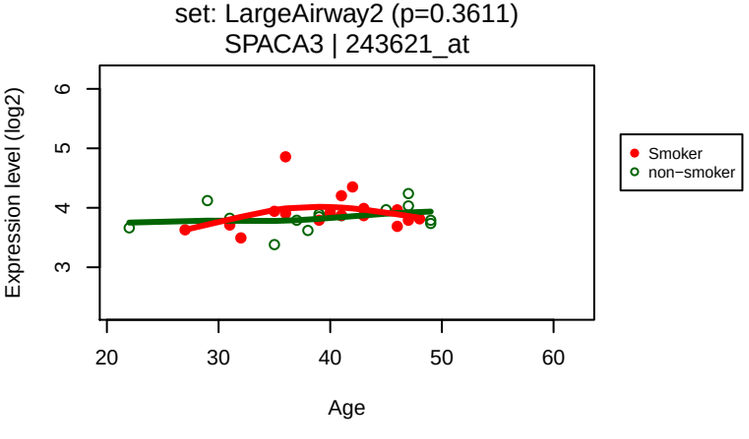

Gene expression ratio (smokers/nonsmokers)
gene: SPACA3 | 243621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0988
p=0.103
p=0.0317
p=0.3094
p=0.3611
p=0.1492
p=0.5087
p=0.3083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

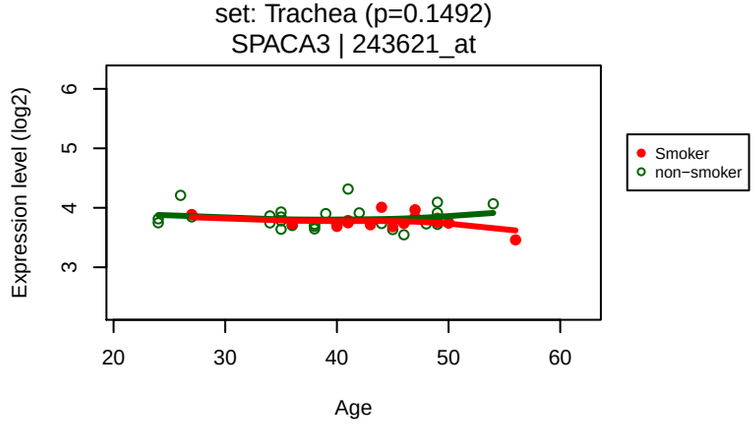

Gene expression ratio in Males
gene: SPACA3 | 243621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.567
p=0.0631
p=0.3855
p=0.3763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPACA3 | 243621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1941
p=0.4634
p=0.1175
p=0.7324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

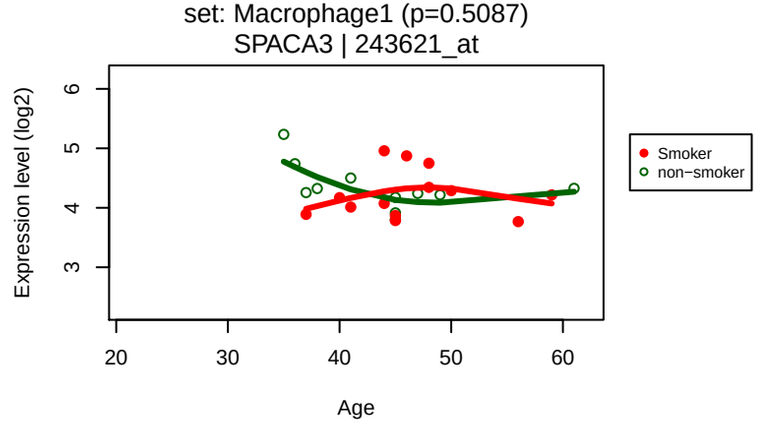

set: SmallAirway1 (p=0.0988)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker

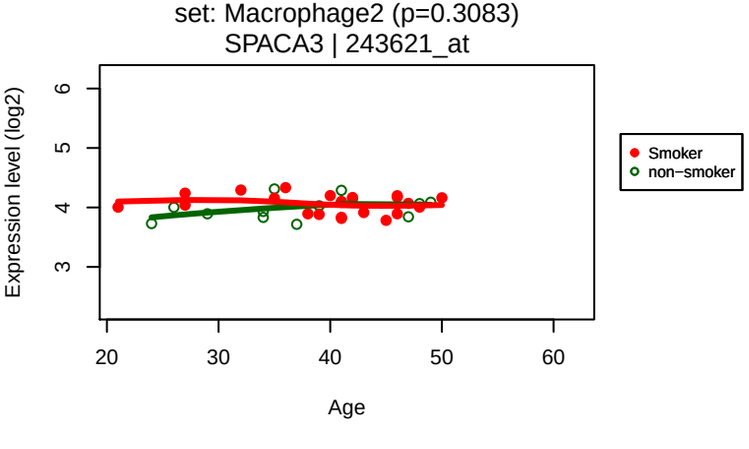

set: SmallAirway2 (p=0.103)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker

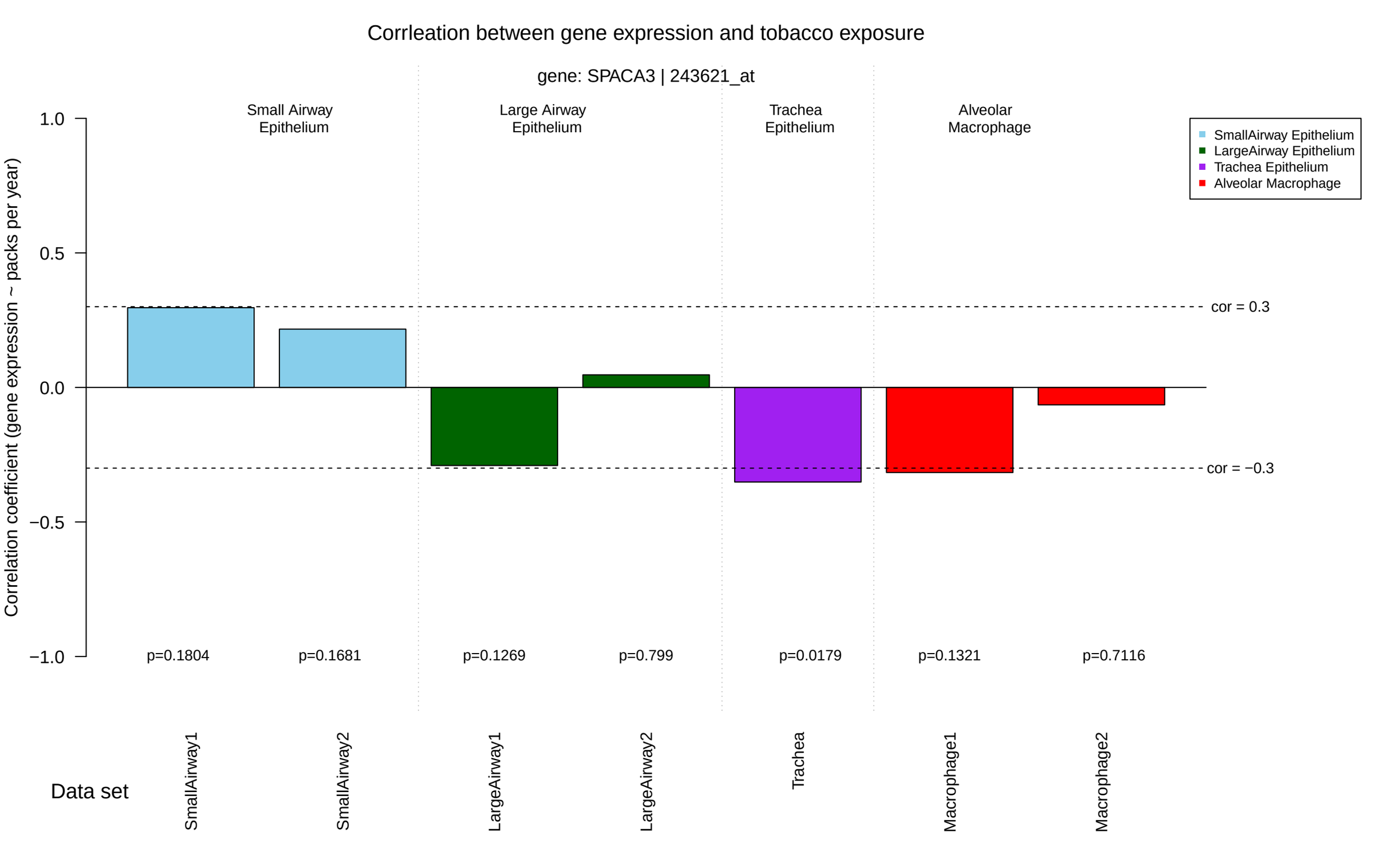

set: LargeAirway1 (p=0.3094)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3611)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1492)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5087)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3083)
SPACA3 | 243621_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPACA3 | 243621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1804
p=0.1681
p=0.1269
p=0.799
p=0.0179
p=0.1321
p=0.7116
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2